Deep Convolutional GAN

for Chest X-Ray Classification

GAN-Based Data Augmentation | 2 Experiments | Pneumonia Detection

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Dataset & Problem Statement

Chest X-Ray Dataset

Split	NORMAL	PNEUMONIA
Train	1,342	3,876
Val	8	8
Test	234	390

Total Training Images: 5,218

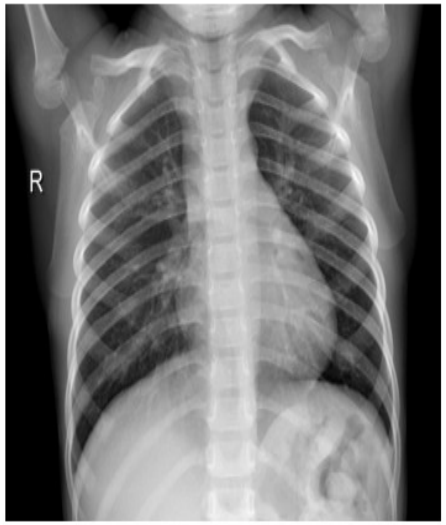
Classes: 2 (NORMAL / PNEUMONIA)

Image Size: 64 × 64 px, Grayscale

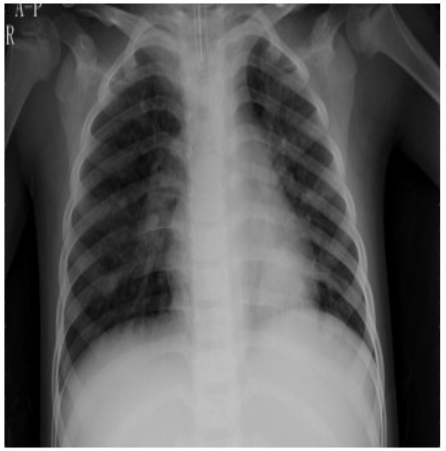
CNN Batch Size: 64 (Exp 2)

CNN Learning Rate: 0.001 → 0.0001

NORMAL



PNEUMONIA



Class Imbalance: 3,876 Pneumonia vs 1,342 Normal → Main reason augmentation is needed

Architecture Overview

Baseline CNN

- 3 × Conv2D (16 → 32 → 64 filters)
- MaxPool after each conv layer
- Dropout = 0.5 (added in Exp 2)
- FC: 4096 → 128 → 2 classes

GAN – Generator

- Input: Random noise $z \in \mathbb{R}^{100}$
- 5 × ConvTranspose2D layers
- BatchNorm + ReLU activations
- Output: 64×64 image (Tanh)

GAN – Discriminator

- Input: 64×64 grayscale image
- 4 × Conv2D layers (32→64→128→256)
- LeakyReLU slope = 0.2
- Output: Real / Fake (Sigmoid)

Loss: CrossEntropyLoss (CNN) | BCELoss (GAN) | Optimizer: Adam | LR: 0.001 (CNN) / 0.0002 (GAN)

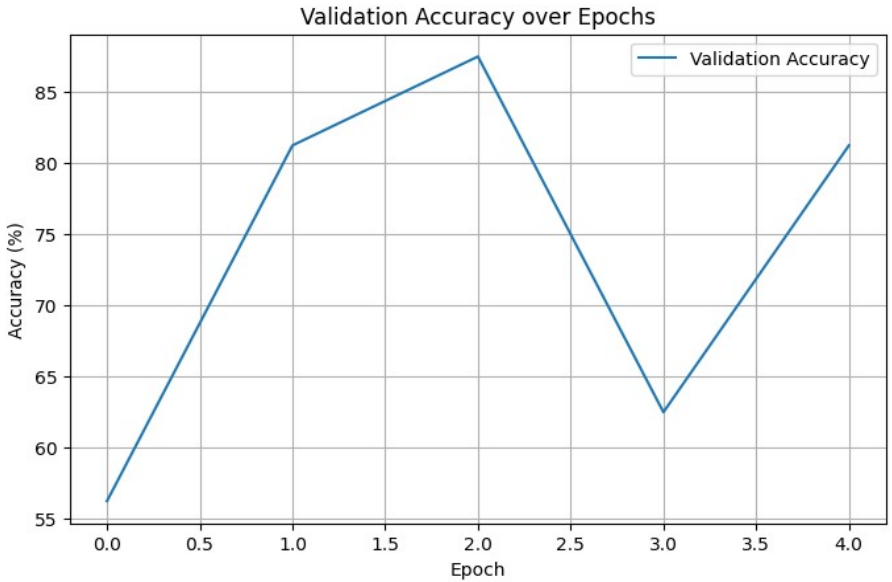
Parameter Variation — Experiment 1 vs Experiment 2

Parameter	Experiment 1 (Original)	Experiment 2 (Improved)
GAN Training Epochs	20	50
GAN Batch Size	64	32
GAN Learning Rate	0.0002	0.0002 (kept same)
Latent Dimension z	100	100 (kept same)
Synthetic Images Generated	500	2,000
Added To Class	PNEUMONIA ❌	NORMAL ✅
CNN Epochs	5	10
CNN Batch Size	32	64
CNN Learning Rate	0.001	0.0001
Dropout	None	0.5
Class Weights	None	Normal × 3.0
Data Augmentation	None	Flip + Rotation 10°

Experiment 1 — Baseline CNN (5 Epochs, LR 0.001, Batch 32)

82.85%	0.0575	87.50%	0.98	0.57
Test Accuracy	Final Train Loss	Peak Val Accuracy	Pneumonia Recall	Normal Recall

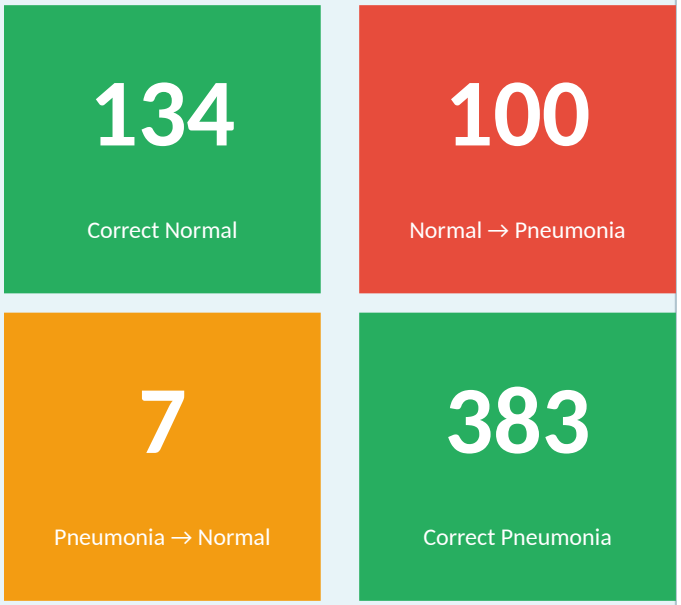




Train Loss: 0.1130 → 0.0575 | Val Accuracy: 50% – 87.5% | Val loss very unstable

Experiment 1 — Baseline CNN Confusion Matrix & Report

Confusion Matrix



Classification Report

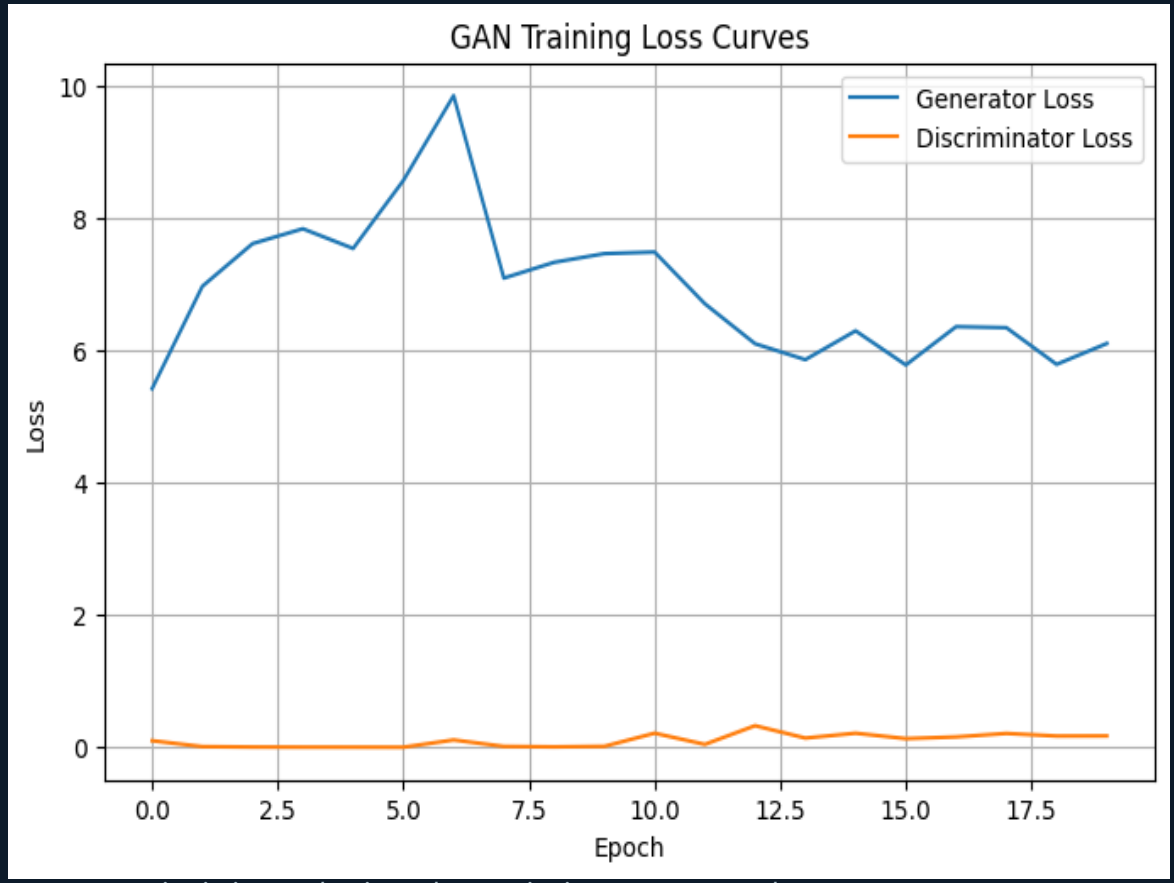
Class	Precision	Recall	F1
NORMAL	0.95	0.57	0.71
PNEUMONIA	0.79	0.98	0.88
Accuracy			82.85%
Macro Avg	0.87	0.78	0.79

Normal recall = 0.57 → 100 normal cases missed
Pneumonia recall = 0.98 → model biased to pneumonia
Test Accuracy: 82.85%

Experiment 1 — GAN Training (20 Epochs)

GAN Config — Exp 1

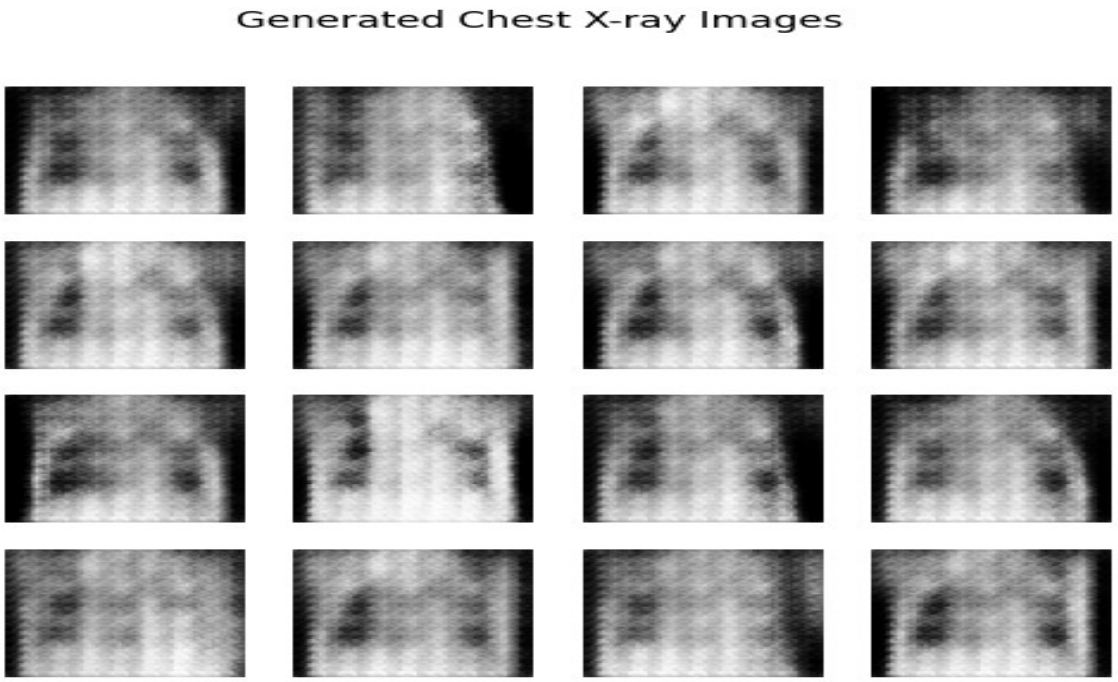
- **Epochs:** 20
- **Batch Size:** 64
- **LR:** 0.0002
- **Latent z:** 100
- **Images Saved:** 500
- **Added To:** PNEUMONIA ❌
- **Loss:** BCELoss



G Loss ~6–10 | D Loss near 0 → Discriminator dominated → Noisy images generated

Experiment 1 — GAN Generated Images (20 Epochs)

16 Synthetic X-rays from GAN trained for 20 epochs | 64×64 px | Grayscale

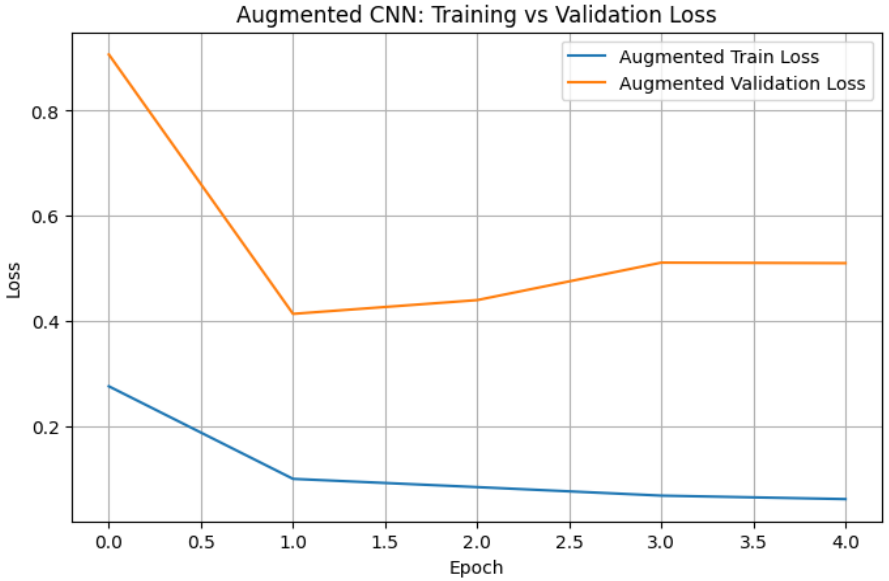


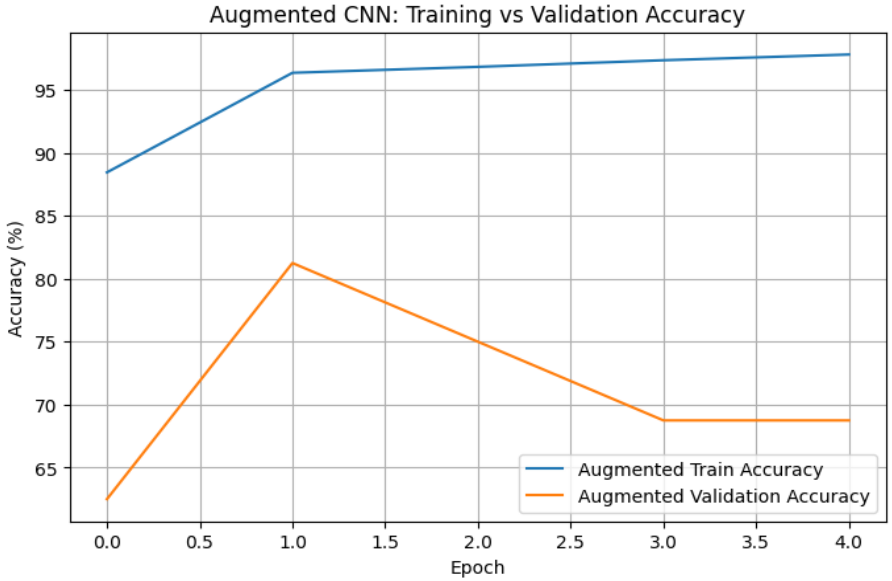
⚠️ Very noisy — lung structure barely visible

✓ Added 500 fake images → PNEUMONIA class

Experiment 1 — Augmented CNN Results (5 Epochs)

75.00%	0.0600	98%	0.99	0.34
Test Accuracy	Final Train Loss	Train Accuracy	Pneumonia Recall	Normal Recall

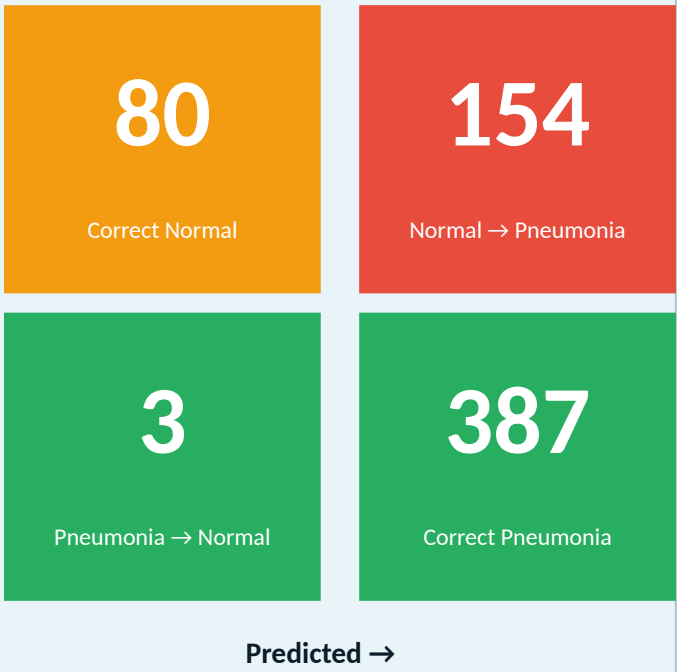




Train Acc 98% | Val Acc dropped to 69% | Accuracy fell from 82.85% → 75% | Severe overfitting

Experiment 1 — Augmented CNN Confusion Matrix & Report

Augmented Confusion Matrix



Classification Report

Class	Precision	Recall	F1
NORMAL	0.96	0.34	0.51
PNEUMONIA	0.72	0.99	0.83
Accuracy			75.00%
Macro Avg	0.84	0.67	0.67

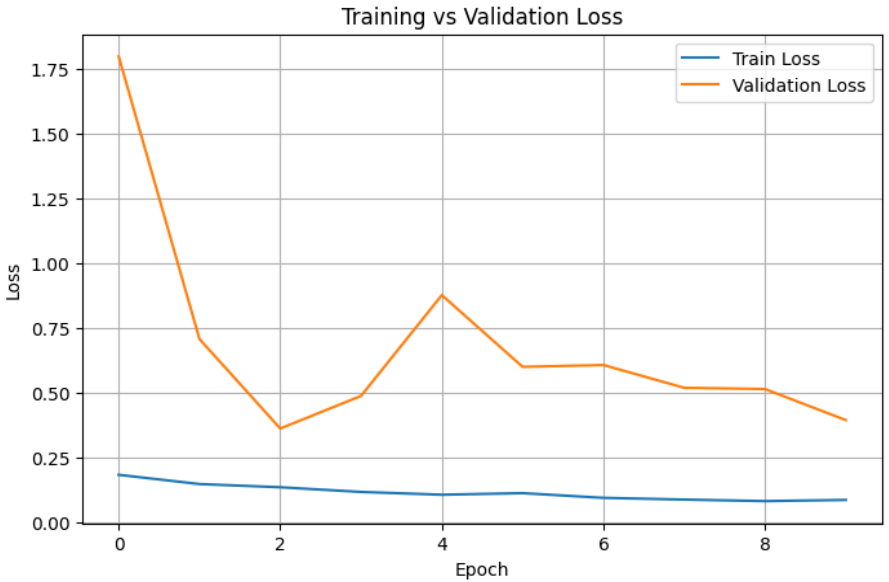
Normal recall dropped: 0.57 → 0.34
154 normal cases wrongly called pneumonia
Fake images added to PNEUMONIA → worsened imbalance
5,216 → 5,716 images | Accuracy: 82.85% → 75%

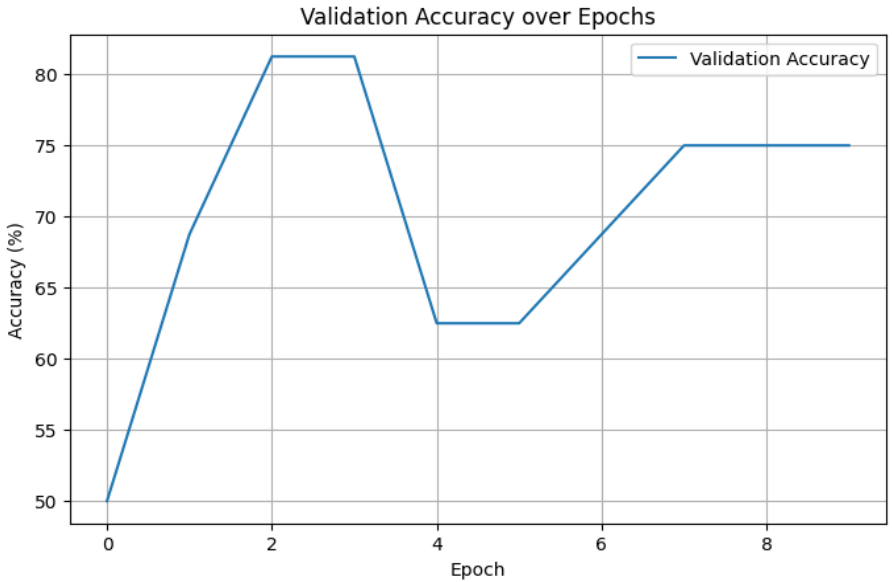
What Went Wrong in Exp 1 → Changes Made for Exp 2

✗ Added 500 fake images to PNEUMONIA class	✓ Added 2,000 fake images to NORMAL class
✗ GAN only trained for 20 epochs → noisy images	✓ Increased GAN to 50 epochs → clearer images
✗ CNN trained for 5 epochs — not enough	✓ Increased CNN to 10 epochs
✗ No class weights — model ignored NORMAL	✓ Added class weights: Normal × 3.0
✗ No Dropout — model memorized training data	✓ Added Dropout = 0.5 to reduce overfitting
✗ Learning rate 0.001 — val loss jumping wildly	✓ Reduced CNN LR to 0.0001 for stability

Experiment 2 — Baseline CNN (10 Epochs, LR 0.001, Batch 64, Dropout 0.5)

73.56%	0.0865	81.25%	0.99	0.32
Test Accuracy	Final Train Loss	Peak Val Accuracy	Pneumonia Recall	Normal Recall

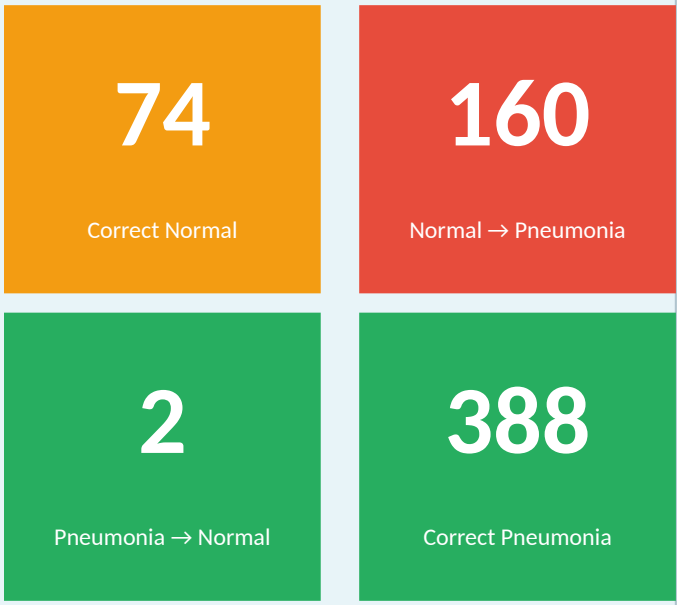




Train Loss: 0.1834 → 0.0865 | Val Accuracy: 50% → 75% | More stable than Exp 1

Experiment 2 — Baseline CNN Confusion Matrix & Report

Confusion Matrix



Classification Report

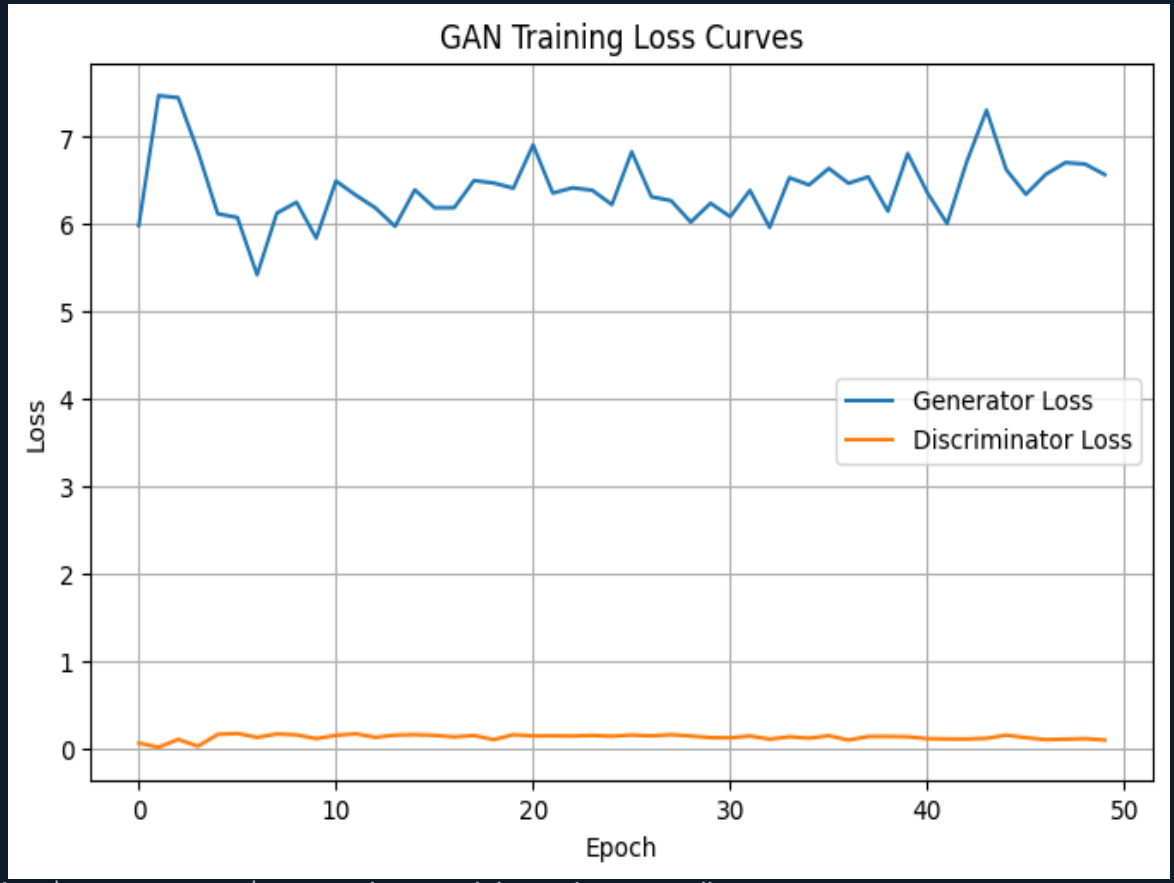
Class	Precision	Recall	F1
NORMAL	0.97	0.32	0.48
PNEUMONIA	0.71	0.99	0.83
Accuracy			73.56%
Macro Avg	0.84	0.65	0.65

Normal recall = 0.32 → 160 normal cases missed
Still biased toward pneumonia — class imbalance effect
This baseline will be improved by augmentation in Exp 2

Experiment 2 — GAN Training (50 Epochs)

GAN Config — Exp 2

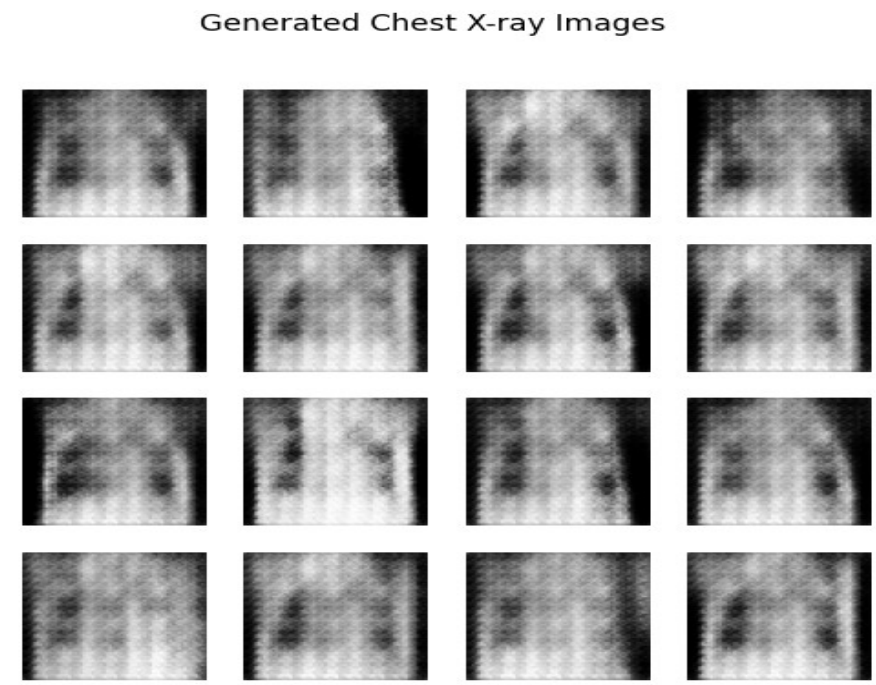
- **Epochs:** 50
- **Batch Size:** 32
- **LR:** 0.0002
- **Latent z:** 100
- **Images Saved:** 2,000
- **Added To:** NORMAL ✅
- **Loss:** BCELoss



G Loss stabilized ~6–7 | D Loss near 0 | 50 epochs = much better image quality

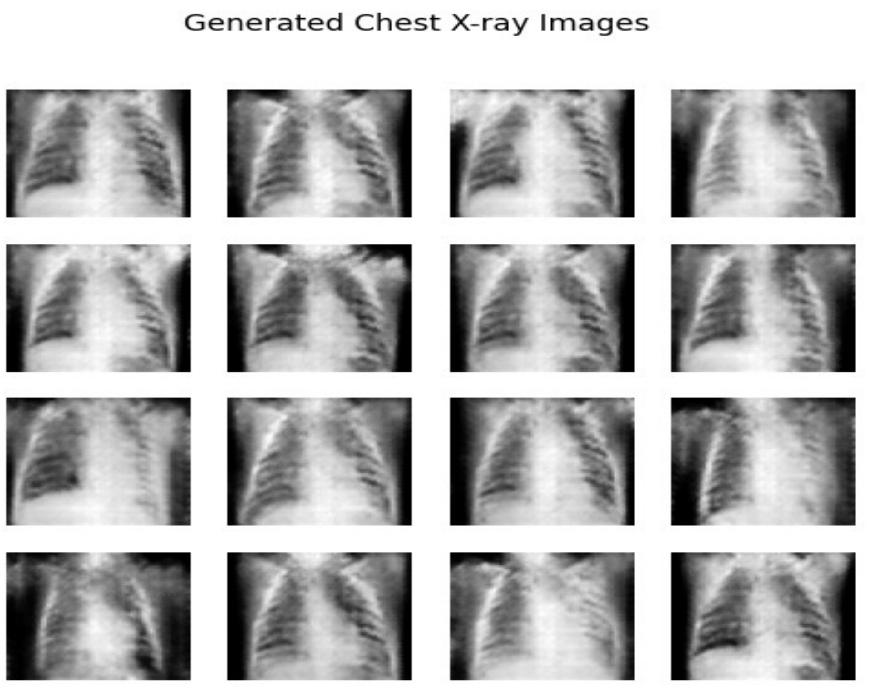
GAN Generated Images — 20 Epochs vs 50 Epochs

Experiment 1 — 20 Epochs



Noisy | No clear lung structure | 500 images

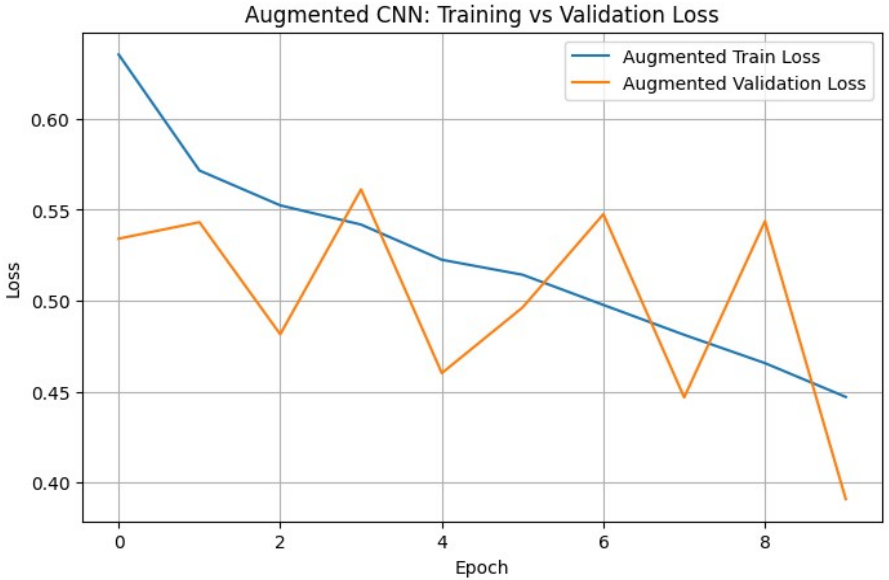
Experiment 2 — 50 Epochs

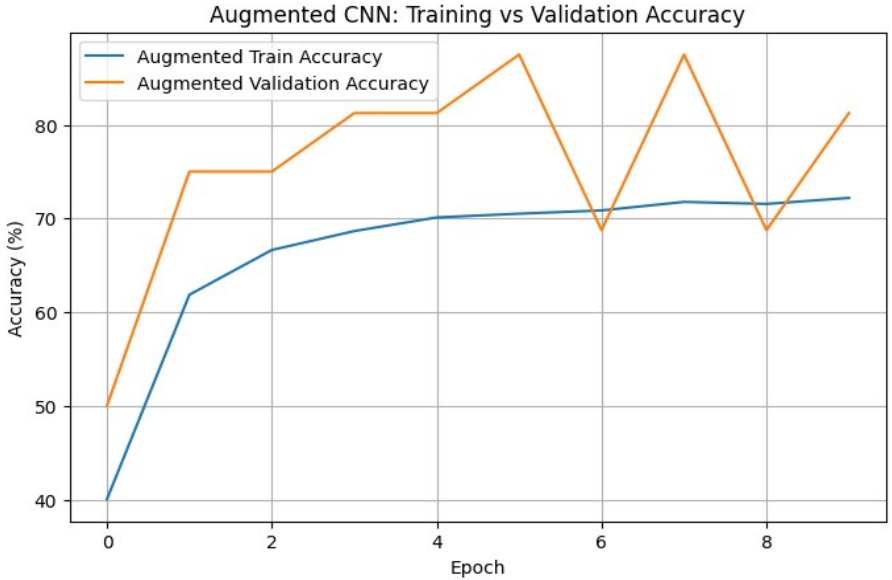


Clear lung shape | Chest structure visible | 2,000 images

Experiment 2 — Augmented CNN Results (10 Epochs, LR 0.0001, Class Weights)

82.21%	0.4471	87.50%	0.87	0.76
Test Accuracy	Final Train Loss	Peak Val Accuracy	Pneumonia Recall	Normal Recall

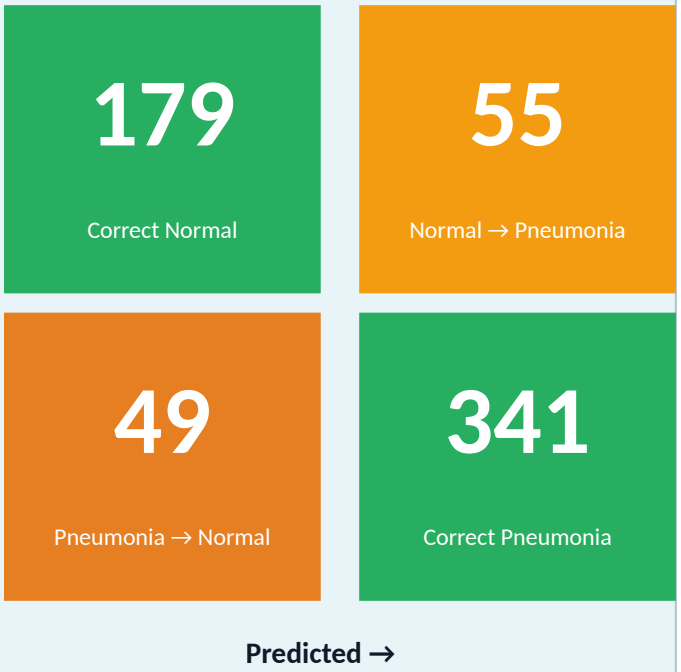




Train Loss: 0.6354 → 0.4471 | Val Accuracy peaked 87.5% | Train & Val both improving — less overfitting

Experiment 2 — Augmented CNN Confusion Matrix & Report

Augmented Confusion Matrix



Classification Report

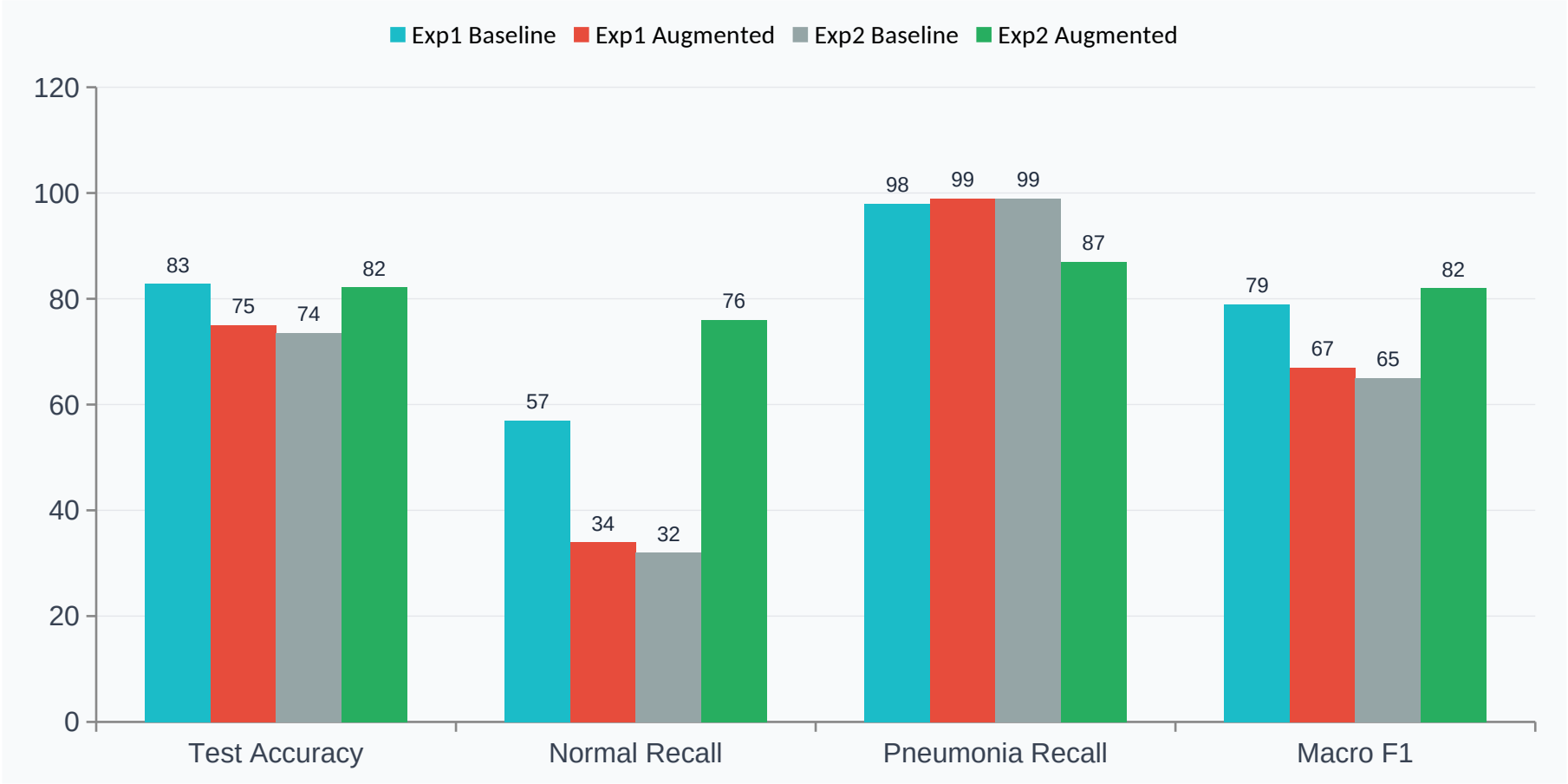
Class	Precision	Recall	F1
NORMAL	0.79	0.76	0.77
PNEUMONIA	0.86	0.87	0.87
Accuracy			82.21%
Macro F1			0.82

Normal recall jumped: 0.32 → 0.76 ✅
Model now identifies normal cases much better
Dataset: 1,342+2,000=3,342 Normal | 3,876+2,000=5,876 Pneumonia
Total training images: 9,218 | Accuracy: 73.56% → 82.21% ✅

Full Comparison — All 3 Models Across Both Experiments

Metric	Exp 1 Baseline	Exp 1 Augmented	Exp 2 Baseline	Exp 2 Augmented ✅
Test Accuracy	82.85%	75.00%	73.56%	82.21%
Training Images	5,216	5,716	5,218	9,218
CNN Epochs	5	5	10	10
GAN Epochs	—	20	—	50
Normal Recall	0.57	0.34	0.32	0.76
Pneumonia Recall	0.98	0.99	0.99	0.87
F1 – Normal	0.71	0.51	0.48	0.77
F1 – Pneumonia	0.88	0.83	0.83	0.87
Macro F1	0.79	0.67	0.65	0.82
Overfitting	Moderate	Severe	Low	Low

Visual Comparison — Accuracy & Recall Across All Models



Final Conclusion

1	Exp 1 augmented PNEUMONIA class → accuracy fell 82.85% → 75% | Normal recall dropped to 0.34
2	Exp 2 fixed the class: added 2,000 images to NORMAL → balanced dataset (3,342 vs 5,876)
3	50 GAN epochs produced clearer chest images vs 20 epochs (visible lung structures)
4	Dropout + Class Weights + LR 0.0001 → reduced overfitting significantly in Exp 2
5	Exp 2 augmented model: accuracy 82.21% | Normal recall 0.76 | Macro F1 = 0.82
→	Future work: Conditional GAN (cGAN) — generate class-specific images for even better balance